

The following questions can help you think critically about your problem-solving processes:

Understanding the Problem

How did you approach understanding the challenge?

The first step I took was to analyze what the program was expected to do for the input value of money in terms of the coins. What I was required to do was to find the minimum number of quarters, dimes, nickels, and pennies.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

Initially, I did not understand how to determine the number of each coin and also track the change left. I came to know that the division would tell me the number of coins in the total amount. After that, I used % to find the change left.

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

Yes, I did break the problem down into smaller parts before writing the code. My plan was to get the quarters first and then use the leftover change to get dimes, nickels, and pennies.

How did you decide on the tools, data structures, or algorithms to use?

The Scanner class was used to take input for the change value from the user. The data type chosen was int because the change value entered by the user was entered in whole cents. The division operator (%) was used to find the coin and change value.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

Rather than writing everything in one sitting, I broke down the coding process into small sections, where I first began with the inputs by the user, followed by the code for each kind of coin, and finally the output statements.

How did you test your solution along the way to make sure it was working?

Different amounts of change were used as input in order to check whether the coins generated by the program are the right ones. For instance, the amount of 87 cents was used to verify whether the answer provided by the program is accurate.

Overcoming Challenges

What part of the problem was the most difficult for you?

The difficult part was figuring out how to calculate the remaining balance for the rest of the coins once each coin was calculated. I had to figure out why there was a need for the % symbol at the end of each line of code.

How did you handle moments when you felt stuck or unsure of what to do next?

If I was confused, I would work out the problem step-by-step with just one coin. With small amounts of change, I could determine what the equations were supposed to do. Through that method, I could determine what the code had to do.

Learning

Was there anything you learned that you think will help you with future challenges?

I got to learn how division and the use of % can help me solve problems involving remainders. Additionally, I learned how dividing problems into simpler components makes programming easier. I will use this knowledge in my future Java applications.